PHASE 6: PROJECT PERFORMANCE TESTING PHASE

Project Details

Particular	Details
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Phase	Project Performance Testing Phase
Date	10 February 2025
Team ID	________________
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6.1 Model Performance Testing

Introduction

The Project Performance Testing Phase was conducted to verify that the Tableau dashboard correctly processed the dataset, rendered all records, implemented preprocessing successfully, and delivered accurate visualizations. The testing also ensured that filters, calculated fields, dashboard layouts, and story navigation functioned efficiently without affecting performance.

Performance Testing Summary

S.No	Parameter	Result
1	Data Rendered	Successfully rendered all 50,000 records and 19 columns
2	Data Preprocessing	Dataset cleaned, validated, standardized, and transformed successfully
3	Utilization of Filters	City filter implemented for dynamic dashboard interaction
4	Calculated Fields Used	Age Group calculated field created and validated
5	Dashboard Design	Interactive dashboard with 9 visualizations developed
6	Story Design	Tableau Story created with 9 analytical scenes

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6.2 Data Rendering & Data Preprocessing

Data Rendered

The Food Ordering Behaviour dataset was imported successfully into Tableau Desktop without any data loss or truncation.

Dataset Statistics

Parameter	Value
Dataset Name	food_ordering_behaviour_dataset.csv
Total Records	50,000
Total Columns	19
Data Loaded	100% Successfully
Data Validation	Passed

A KPI card displaying the **Count of Order ID = 50,000** confirmed that the complete dataset had been loaded correctly and was available for analysis.

Data Preprocessing

Before visualization, the dataset was cleaned and transformed to improve quality and analytical accuracy.

Preprocessing Activities

- Verified dataset structure.
- Checked column names and data types.
- Removed duplicate records.
- Validated missing values.
- Standardized categorical values.
- Verified numerical measures.
- Created calculated fields.
- Prepared the dataset for Tableau visualization.

Outcome

The preprocessing stage ensured that the dataset was consistent, accurate, and visualization-ready.

6.3 Utilization of Filters & Calculated Fields

Dashboard Filters

Interactive filters were implemented to improve dashboard usability and support dynamic analysis.

Implemented Filter

Filter	Purpose
City	Allows users to analyze food ordering behaviour for a selected city

The City filter dynamically updates all connected charts, enabling comparison of cuisine demand, meal preferences, company types, revenue, and delivery trends across different locations.

Calculated Fields

A calculated field was created to categorize customers into meaningful demographic groups.

Age Group Calculation

Condition	Age Group
Age ≤ 20	Child
Age 21–30	Gen Z
Age 31–40	Millennial
Age > 40	Adults

Benefits

- Simplifies customer segmentation.
- Improves dashboard readability.
- Enables age-based comparative analysis.
- Supports business decision-making.

6.4 Dashboard & Story Performance

Dashboard Design Testing

The Tableau dashboard was tested for responsiveness, navigation, and visualization accuracy.

Dashboard Features

- Professional beige/cream theme.
- Responsive layout.
- Interactive City filter.
- Dynamic KPI cards.
- Tooltips for additional insights.
- Consistent color palette.
- Clear chart labeling.

Dashboard Visualizations

A total of **9 visualizations** were successfully integrated.

S.No	Visualization
1	KPI Cards
2	Heat Map
3	Grouped Bar Chart
4	Pie Chart
5	Company Type vs Meal Type Bar Chart
6	Donut Chart
7	Revenue Bar Chart
8	City-wise Order Volume Chart
9	Delivery Time Distribution Chart

Story Design Testing

The Tableau Story was tested to ensure smooth navigation and logical presentation of insights.

Story Scenes

Scene	Description
Scene 1	KPI Overview
Scene 2	Cuisine Demand Across Cities
Scene 3	Age Group Analysis
Scene 4	Company Type vs Meal Type
Scene 5	Customer Rating Analysis
Scene 6	Revenue Contribution
Scene 7	Cuisine Preference Distribution
Scene 8	City-wise Order Volume
Scene 9	Delivery Time Analysis

Each story point successfully displayed the intended visualization and supported a coherent analytical narrative.

Performance Evaluation

Testing Results

Test Case	Expected Result	Actual Result	Status
Dataset Loading	All records loaded	50,000 records loaded	Passed
Data Cleaning	Clean dataset	Successfully cleaned	Passed
Calculated Fields	Correct age categorization	Successfully categorized	Passed
Dashboard Rendering	All charts displayed	9 charts rendered	Passed
Interactive Filters	Dynamic updates	Working correctly	Passed
Story Navigation	Smooth transitions	Working correctly	Passed

Performance Summary

The Tableau solution performed efficiently throughout testing. All records were processed successfully, calculated fields produced accurate results, filters responded dynamically, and dashboards rendered without noticeable delay. The Story module presented insights in a structured sequence, allowing users to explore customer behaviour, cuisine demand, meal preferences, revenue, and delivery performance effectively.

Conclusion

The Project Performance Testing Phase verified that the Food Ordering Behaviour analytics solution met all functional and performance expectations. The system successfully handled the complete dataset of 50,000 records, implemented preprocessing and calculated fields accurately, supported interactive filtering, and delivered a responsive dashboard with nine visualizations and a comprehensive Tableau Story. The successful completion of this phase confirmed that the project was ready for deployment, web integration, and final demonstration.